Bayesian Method

1. A medical test for a certain disease is 95% accurate. If a person has the disease, the test will correctly identify it 95% of the time. However, the test also has a 2% false positive rate. If 1% of the population has the disease, what is the probability that a person actually has the disease given that they tested positive?
   Answer: 0.324 or 32.4%

2. A factory produces 70% of its products from Machine A and 30% from Machine B. Machine A produces 95% non-defective items, while Machine B produces 80% non-defective items. If a randomly selected product is found to be non-defective, what is the probability it was made by Machine A?
   Answer: 0.324 or 32.4%

3. Suppose you roll a fair die and are interested in calculating the posterior probability of each face given that you rolled an even number. What is the posterior probability for each of the even numbers (2, 4, 6)?
   Answer: 1/3 for each even number (2, 4, 6)

4. In a town, 40% of the people read newspapers A, 20% read newspapers B, and 10% read both. If a person is known to read newspaper B, what is the probability that they also read newspaper A?
   Answer: 0.5 or 50%

5. You are working with a coin that may be biased. The prior probability that the coin is fair (P(F)) is 0.5, and the prior probability that the coin is biased (P(B)) is 0.5. If the coin is biased, it lands heads with a probability of 0.8. You flip the coin 10 times and observe 7 heads. What is the posterior probability that the coin is biased?
   Answer: 0.86 or 86%

6. A drug test can correctly identify drug users 99% of the time and non-users 98% of the time. In a certain population, 5% of individuals are drug users. A person is randomly selected from the population and tests positive. What is the probability that the person is a drug user?
   Answer: 0.7174 or 71.74%

7. In a Bayesian A/B test for an e-commerce website, you are trying to decide whether a new feature (B) improves conversion rate compared to the current version (A). The prior belief is that both A and B are equally likely to be better (P(A) = P(B) = 0.5). After running the test, you observe 120 conversions out of 500 users in group A and 150 conversions out of 500 users in Group B. What is the posterior probability that version B is better than version A?
   Answer: 0.915 or 91.5%

8. A factory produces 60% of its goods from machine X and 40% from machine Y. Machine X produces 90% defect-free items, and machine Y produces 80% defect-free items. A random item is selected and found to be defective. What is the probability it was produced by machine Y?
   Answer: 0.4706 or 47.06%

9. A team is trying to predict whether a startup will succeed based on its current growth rate. They estimate that 60% of startups succeed if their growth rate is high, and 30% succeed if

the growth rate is low. The probability of having a high growth rate is 50%. Given that a startup has a high growth rate, what is the probability that it will succeed?
Answer: 0.6 or 60%

Imputation

10. You are given the following dataset with some missing values:
X
12
15
14
NaN
16
NaN
10
Perform mean imputation to fill the missing values.
Answer: Mean = 13.4

11. You are given the following dataset:
X
5
7
NaN
8
NaN
9
6
Use median imputation to fill in the missing values.
Answer: Median = 7

12. The dataset below has some missing values:
X
18
19
NaN
20
18
NaN
21
Fill in the missing values using mode imputation.
Answer: Mode = 18
13. You are working with a dataset that has the following data:
Age	Income	City
25	NaN	New York
NaN	40000	Chicago

35	60000	NaN
40	NaN	New York
29	50000	Chicago
32	45000	Chicago

Apply mean imputation for the missing values in the 'Age' and 'Income' columns, and mode imputation for the 'City' column.
Answer:
Mean for Age = 32.2
Mean for Income = 48750
Mode for City = Chicago

14. Consider the following dataset with missing values:
X1	X2	X3
12	NaN	45
13	56	NaN
NaN	50	42
14	NaN	41
13	52	44
Perform median imputation for columns X1, X2, and X3.
Answer:
Median for X1 = 13
Median for X2 = 52
Median for X3 = 43
Updated Dataset:

KNN Imputer

15. Consider the following dataset with missing values:

Age	Income	Education Level
25	50,000	1
30	60,000	2
28	NaN	1
NaN	45,000	3
40	70,000	2

Using KNN Imputer with k=2, fill in the missing values for Age and Income.

Answer:

Age	Income	Education Level
25	50,000	1
30	60,000	2
28	52,500	1

| 32.5 | 45,000 | 3 |
| 40 | 70,000 | 2 |

16. You are given a dataset:

Feature A	Feature B	Feature C
5	NaN	10
6	20	9
7	NaN	8
8	18	NaN
NaN	17	12

Apply KNN Imputer with k=3 to fill in the missing values.

Answer:

Feature A	Feature B	Feature C
5	19	10
6	20	9
7	19	8
8	18	9.67
6.33	17	12

17. Given the following dataset with missing values:

X1	X2	X3	X4
1.0	NaN	3.0	2.0
2.0	1.0	NaN	3.0
NaN	2.0	2.0	1.0
3.0	3.0	4.0	NaN
4.0	NaN	1.0	4.0

Using KNN Imputer with k=2, impute the missing values.

Answer:

X1	X2	X3	X4
1.0	1.5	3.0	2.0
2.0	1.0	3.0	3.0
2.5	2.0	2.0	1.0
3.0	3.0	4.0	2.5
4.0	2.0	1.0	4.0

18. Suppose you have the following incomplete dataset:

| Height | Weight | Age | Salary |

Height	Weight	Age	Salary
180	75	30	NaN
170	NaN	25	60,000
160	60	NaN	50,000
150	50	35	40,000
NaN	70	28	55,000

If you apply KNN Imputer with k=3, describe how the missing values would be imputed.

Answer:

Height	Weight	Age	Salary
180	75	30	51,666.67
170	62.5	25	60,000
160	60	29	50,000
150	50	35	40,000
165	70	28	55,000

19. You are tasked with imputing missing values in the following dataset:

Temperature	Humidity	Wind Speed	Air Pressure
20	NaN	12	1012
25	60	NaN	1008
NaN	65	10	NaN
22	70	NaN	1010
18	NaN	14	1005

Using KNN Imputer with k=3 and Euclidean distance, explain the procedure for filling in the missing values.

Answer:

Temperature	Humidity	Wind Speed	Air Pressure
20	65	12	1012
25	60	12	1008
22.33	65	10	1009
22	70	12	1010
18	65	14	1005

20. Consider a dataset:

Feature A	Feature B	Feature C
10	20	NaN
15	NaN	5
NaN	25	8
12	NaN	6

You are using KNN Imputer with k=2. How would the missing values be filled?

Answer:

Feature A	Feature B	Feature C
10	20	6.5
15	22.5	5
13.5	25	8
12	22.5	6

These are the questions with the final answers based on the KNN Imputation process.

Z-Score

21. The heights of a group of people are normally distributed with a mean of 160 cm and a standard deviation of 10 cm. What is the Z-score of a person whose height is 175 cm?
    Answer: 1.5

22. A test score is 70, with a mean score of 50 and a standard deviation of 10. What is the Z-score?
    Answer: 2

23. The average weight of a population is 65 kg, with a standard deviation of 5 kg. What is the Z-score for a person weighing 72 kg?
    Answer: 1.4

24. A student's score on an exam is 80, with a class mean of 70 and a standard deviation of 8. What is the Z-score of the student? What percentage of students scored lower than this student?
    Answer: Z-score = 1.25; Percent = 89.44%

25. The average age of employees at a company is 35 years with a standard deviation of 7 years. What is the Z-score for an employee who is 45 years old? What percentage of employees are younger than this employee?
    Answer: Z-score = 1.43; Percent = 92.36%

26. The weights of students are normally distributed with a mean of 60 kg and a standard deviation of 12 kg. What is the Z-score for a student weighing 45 kg?
    Answer: -1.25

27. A test has a mean of 50 and a standard deviation of 5. If a student scored 65, what is their Z-score?
    Answer: 3

28. The mean IQ of a population is 100, with a standard deviation of 15. What is the Z-score for an individual with an IQ of 130?
    Answer: 2

IQR (Interquartile Range)

29. Given a dataset with a first quartile (Q1) of 20 and a third quartile (Q3) of 40, calculate the interquartile range (IQR).
    Answer: 20

30. The Q1 of a dataset is 15, and the Q3 is 35. What is the interquartile range?
    Answer: 20

31. In a dataset, Q1 = 10 and Q3 = 25. Find the IQR.
    Answer: 15

32. In a dataset, the first quartile is 12, the third quartile is 28, and the median is 20. Find the IQR. Is a value of 5 an outlier?
    Answer: IQR = 16; No

33. A dataset has the following five-number summary: Minimum = 5, Q1 = 12, Median = 18, Q3 = 25, and Maximum = 35. Calculate the IQR. Is an observation of 40 an outlier?
    Answer: IQR = 13; No

34. The Q1 of a dataset is 14, and the Q3 is 30. Calculate the IQR and determine if a value of 5 is an outlier.
    Answer: IQR = 16; No

35. For a dataset, Q1 = 22 and Q3 = 48. Calculate the IQR.
    Answer: 26

36. The first quartile (Q1) is 18 and the third quartile (Q3) is 35 in a dataset. Find the IQR.
    Answer: 17

Logistic Regression

37. A logistic regression model is fitted on a dataset with one predictor variable. The logistic regression equation is given by:
    $$\text{logit}(p) = -2 + 0.8X$$
    Where p is the probability of success and X is the predictor. If X=3, find the probability of success.
    Answer: 0.599.

38. A logistic regression model is fitted with a single predictor variable. The logistic regression equation is:
    $$\text{logit}(p) = 1.5 + 0.4X$$

Where P is the probability of success and X is the predictor. If X=2, find the probability of success.

Answer: 0.909.

39. Given a logistic regression model:

$$\text{logit}(p)=-1+0.6X$$

Where P is the probability of success, find the probability of success when X=5

Answer: 0.881.

40. A logistic regression model is given as:

$$\text{logit}(p)=0.5-0.3X$$

If X=4, what is the probability of success?

Answer: 0.332

41. The logistic regression model for a binary classification problem is given by:

$$\text{logit}(p)=0.2+0.5X1-0.3X2$$

Where P is the probability of success, X1 and X2 are predictor variables. Find the probability of success when X1=4 and X2=2.

Answer: 0.832

42. A logistic regression model is given by:

$$\text{logit}(p)=-1.2+0.4X1+0.6X2$$

Where P is the probability of success, X1 and X2 are predictor variables. Find the probability of success when X1=5 and X2=3.

Answer: 0.931

43. The logistic regression model for a classification problem is given by:

$$\text{logit}(p)=-0.8+0.7X1-0.4X2$$

Where P is the probability of success. Find the probability of success when X1=3 and X2=1.5

Answer: 0.668

44. Given a logistic regression model:

$$\text{logit}(p)=1.2-0.5X1+0.8X2$$

Where P is the probability of success, calculate the probability when X1=4 and X2=2.

Answer: 0.69

45. A logistic regression model is fitted with two predictors:

$$\text{logit}(p)=-0.9+0.3X1+0.5X2$$

Find the probability of success when X1=6X_1 = 6X1=6 and X2=2.

Answer: 0.87

K-Nearest Neighbour

46. You are given a dataset with three features: Height (cm), Weight (kg), and Age (years) of three individuals along with their corresponding label, which indicates whether they are athletes or not (1 for athlete, 0 for non-athlete):

Height (cm)	Weight (kg)	Age (years)	Label
170	65	25	1
180	85	28	1
160	50	23	0

Now, a new individual with Height = 175 cm, Weight = 75 kg, and Age = 27 years is introduced. Use the K-nearest neighbours algorithm (K=1 and K=3) to classify this new individual. Use Euclidean distance as the metric.
Answer: for K=1, the new individual is classified as athlete (1).
for K=3, the new individual is classified as athlete (1).

47. Consider a dataset with the following points and their labels. The features are X1 and X2, and the labels are either 0 or 1.

X1	X2	Label
2	4	0
4	6	1
5	7	1
3	3	0

Now, you are given a new point (3, 5). Use K=2 in the K-nearest neighbors algorithm to classify the new point, using Euclidean distance.
Answer: for K=2, the new point can be classified as either 0 or 1.

48. You are given a dataset with three features: Length (cm), Width (cm), and Weight (grams) of three objects, along with their corresponding label, which indicates whether they belong to Category A (label 1) or Category B (label 0).

Length (cm)	Width (cm)	Weight (grams)	Label
10	5	100	1
15	7	200	0
12	6	150	1

Now, a new object with Length = 13 cm, Width = 6 cm, and Weight = 160 grams is introduced. Use the K-nearest neighbors algorithm with K=2 to classify this new object using Euclidean distance.
Answer: Category A (label 1).

49. You have a dataset of customers' income and spending score from a mall, and their loyalty label (1 for loyal, 0 for not loyal).

Income (in $1000)	Spending Score	Label
40	50	1
50	30	0
60	40	1
30	20	0

A new customer has an income of $45,000 and a spending score of 35. Use K=3 to classify the new customer as loyal (1) or not loyal (0) using Euclidean distance.
Answer: for K=3, the new customer is classified as loyal (label 1).

50. You have the following data of flower characteristics in a garden:

Sepal Length (cm)	Sepal Width (cm)	Label
5.1	3.5	0
4.9	3.0	0
6.2	2.9	1
5.8	2.7	1

A new flower is observed with a sepal length of 5.5 cm and sepal width of 3.2 cm. Use K=1 and K=3 to classify the new flower using Euclidean distance.
Answer: for K=3, the new flower is classified as label 0**.**